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Convolution dictionary learning for visible-infrared image fusion via local processing

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Abstract

Although convolutional sparse coding overcomes the limitations of block-based sparse representation in the process of convolutional dictionary learning, it relies too much on the Alternating Direction Method of Multipliers(ADMM) formula in the Fourier domain. This leads to a loss of locality, impacting the quality of the fused image which tends to have a fuzzy texture. To solve this problem, we use a local processing convolution dictionary-learning method to obtain a dictionary and apply the obtained dictionary to the fusion of visible-infrared images. The proposed method not only solves the problem of global convolutional sparse coding, but also overcomes the blurry fusion image defect. Experimental results show that the proposed method is superior to the comparison methods in subjective and objective evaluation. Compared with deep learning fusion methods, our fusion framework achieves an average improvement of 5.18%, 5.02%, and 4.77% in the objective evaluation QAB/F, Qe and Qp, respectively.

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Keywords: Visible and infrared image fusion; convolution sparse representation; convolution dictionary learning; local-global

1. Introduction

A thermal infrared image can natively distinguish the background from the monitoring target, to good effect in all-weather and night monitoring. However, the obtained visible-light image has low clarity, is blurry, and achieves a very low degree of recognition. The image can provide texture details with high spatial resolution and clarity, but at night or in the presence of obstructions the target cannot be well monitored. Fusion technology holds the potential

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to not only achieve high resolution in visible light, but also provide thermal radiation information from thermal infrared images.

Contemporary infrared-visible fusion algorithms have developed rapidly, and scholars at home and abroad have proposed many novel and effective fusion algorithms[1-5]. Currently, fusion methods such as multi-scale transformation, sparse domain, and deep learning are the most popular. Image-fusion methods are based on multi-scale transformation, and assume that the image will be represented by layers in different particles. These methods decompose the source image into multiple levels, use specific rules to fuse specific layers, and reconstruct the target image accordingly. The sparse domain-based method uses a representation of an image which has a linear combination of sparse basis in a super-complete dictionary, the key to its good performance. Deep learning-based fusion leverages the superior power of various neural networks providing this method with strong adaptability and anti-noise capacity. Nevertheless, all of the above-mentioned popular fusion algorithms have some drawbacks. Although multi-scale transformation-based methods rely excessively on predefined transformations with corresponding decomposition and reconstruction levels, there is currently no standard to facilitate simple evaluation of these transformations and levels. Although it is easy to choose the conversion to reduce the fusion performance, it remains difficult to build a suitable super-complete dictionary with a good representation of the target data based on the traditional sparse domain method; the fusion method based on deep learning needs to incorporate an appropriate neural network and adjust the corresponding parameters.

This paper proposes infrared-visible image fusion method via local processing convolution dictionary learning. The low-pass decomposition method is used to decompose the source image into low-pass components and detail content. We use the maximum value strategy to obtain the fused low-pass component. For the detail component, we use the learned local processing convolution dictionary and convolution sparse coding, and then reconstruct the low-pass component and details to obtain the final fusion image.

2. Convolution dictionary learning based on local processing

Convolutional sparse coding is an increasingly popular model in the field of signal and image processing because it overcomes some of the limitations of traditional block-based sparse representations. Although many researchers have solved the problem of dictionary learning under this model, their studies have relied on the ADMM formula in the Fourier domain, which results in loss of locality and the relationship with traditional block-based sparse tracking. Recently, V. Pappyan proposed a new matching tracking model based on Toeplitz transform[6]. The proposed model transcends the computational complexity limitation posed by fast Fourier transforms. In this study, we use V. Pappyan's slice-based convolution dictionary learning method. After applying this learning method to the USC-SIPI image dataset [1,2](50 standard images of size 512×512), we obtain a dictionary filtering size of $11 \times 11 \times 100$.

3. Fusion method

The fusion method proposed in this paper, using learned dictionary filtering and convolutional sparse coding, is shown in Fig.1. It includes four major steps: source image $\{A, B\}$ decomposition; low-pass component $\{L_A, L_B\}$ fusion; detail component $\{H_A, H_B\}$ fusion; and finally, image reconstruction.

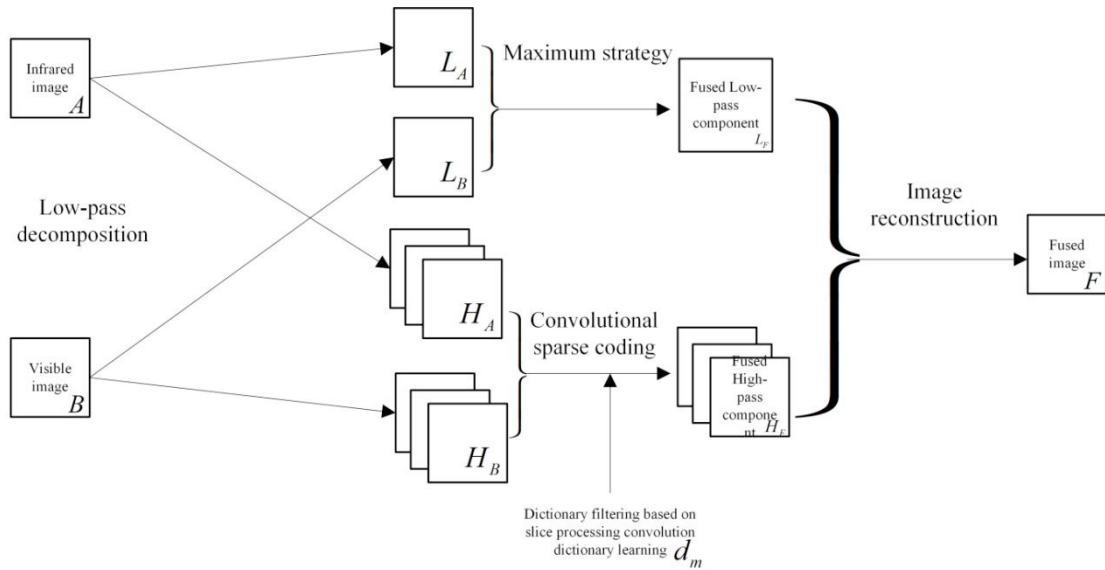


Fig. 1. Infrared-visible image fusion framework.

4. Experiment

Fig.2 shows the images of the four infrared and visible-light sources tested, including complex environments with features such as pedestrians, vehicles, bushes, houses, etc. at night.

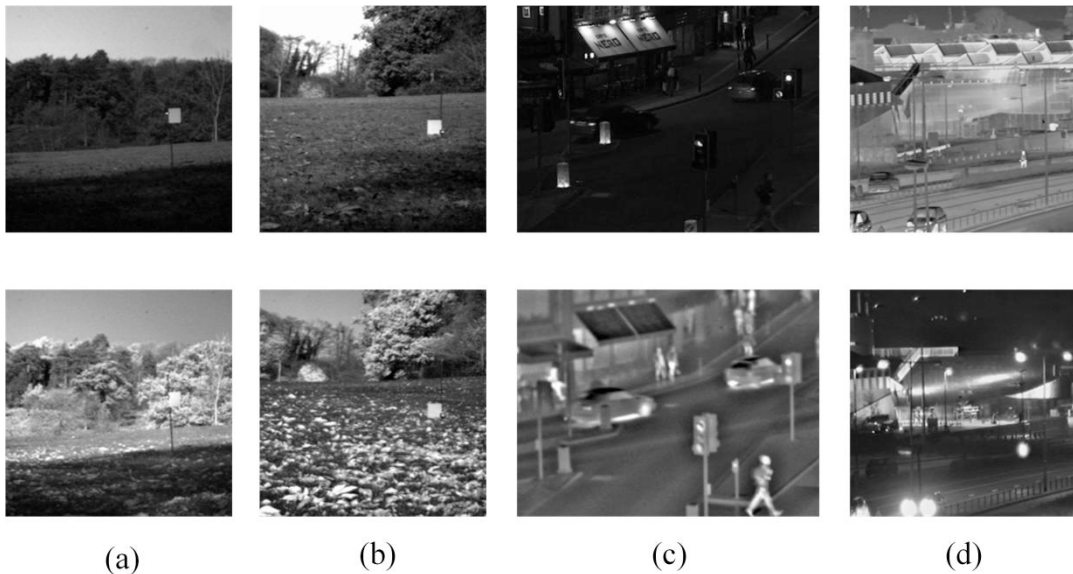


Fig. 2. Infrared and visible source images.

(1) Fusion image result

Figs.3-6 show the fusion results of eight methods applied to four groups of infrared-visible images: (a) indicates infrared source image; (b) indicates visible source image; (c) represents results obtained using the NSCT-SSIM fusion algorithm proposed by Zhang Lei[7]; (d) represents results obtained using the GTF fusion algorithm proposed

by Ma Jiayi[8]; (e) represents results obtained using the MSVD fusion proposed by VPS Naidu[9]; (f) represents results obtained using the JSRSD fusion algorithm proposed by Liu Chunhui[10]; (g) represents results obtained using the latrr fusion algorithm proposed by Li Hui[11]; (h) represents results obtained using the deep learning fusion algorithm proposed by Li Hui[12]; (i) represents results obtained using the GFCE fusion algorithm proposed by Zhou Zhiqiang[13]; and finally, (j) represents results obtained using the proposed fusion algorithm. It can be seen from four sets of fusion results that image obtained by GTF is of the worst quality, and that the visible-light information in the fusion result is too heavy. The fusion image obtained by MSVD is slightly better than that obtained from GTF, but some details are blurred (as shown in Figure 3 (e), the stone is blurred, and the overall image is dark). There is too little visible-light information in the JSRSD and GFCE fusion images, resulting in blurred image details. The NSCT-SSIM and latrr fusion results are superior to those of the first four methods, but compared to deep learning and our proposed method, the overall image tone is low, and there is too much visible-light information in the fused image. Overall, our algorithm produces results similar to those of the deep learning method, but richer in detail and thus of higher quality.

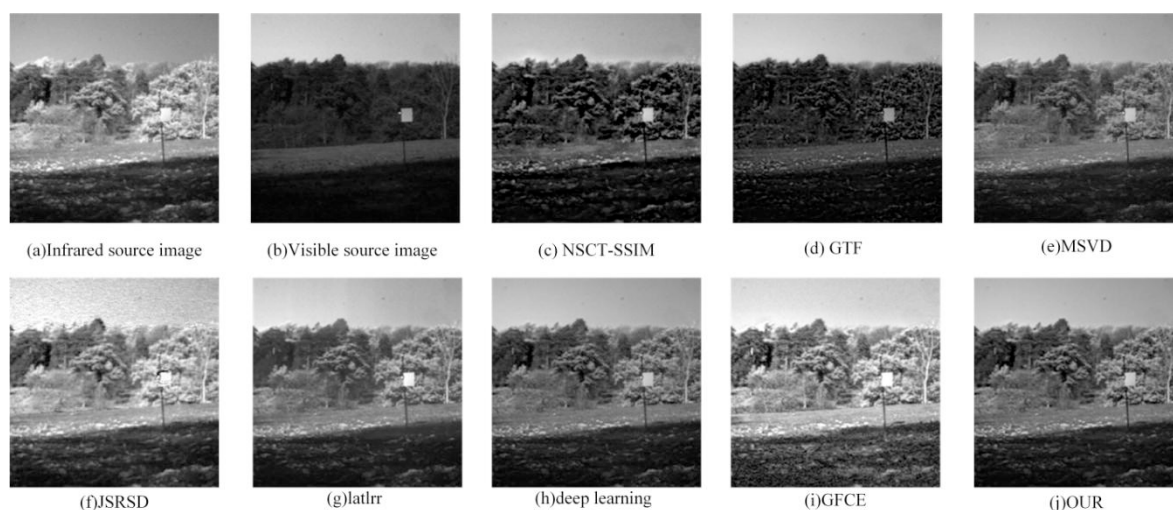


Fig. 3. Fused results of (a) test images.

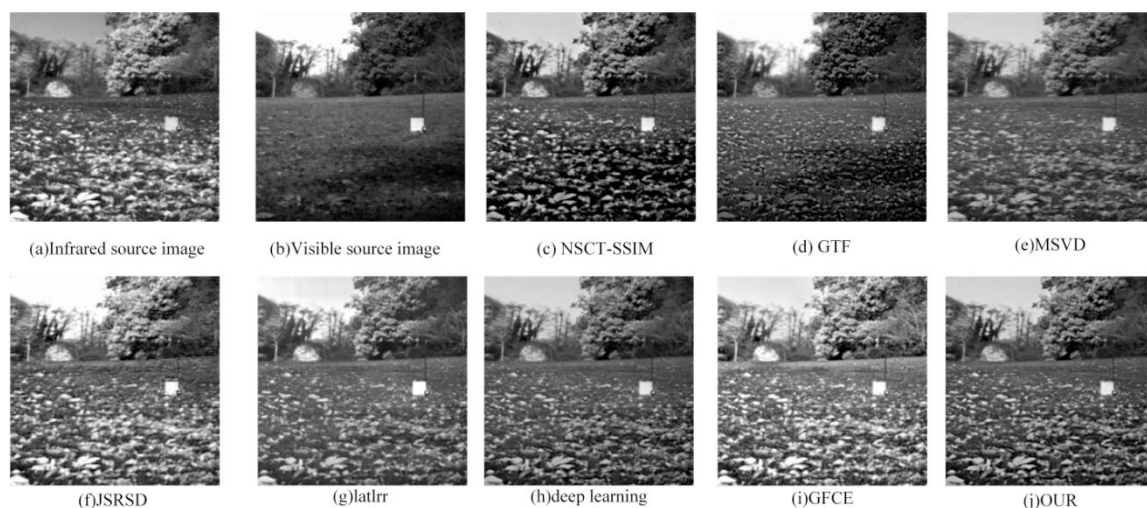


Fig. 4. Fused results of (b) test images.

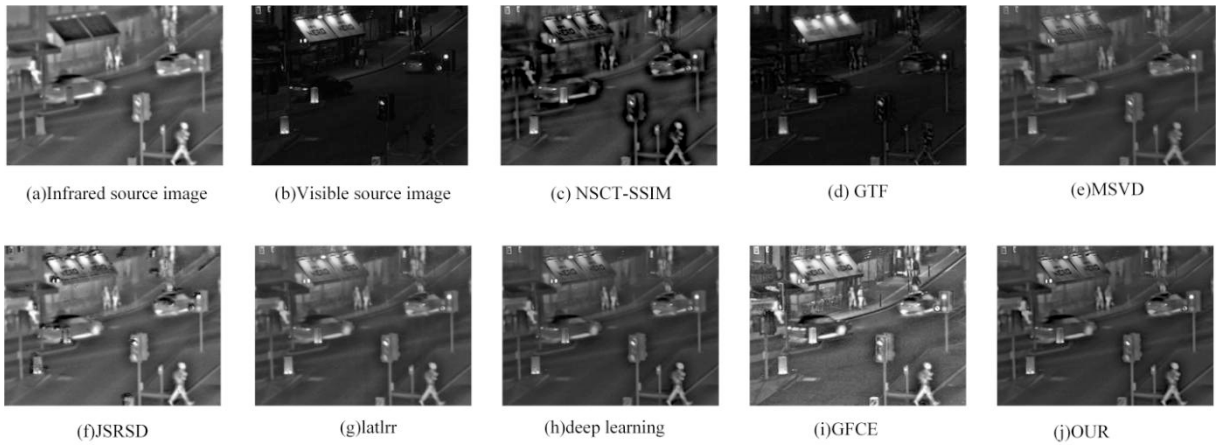


Fig. 5. Fused images of (c) test image.

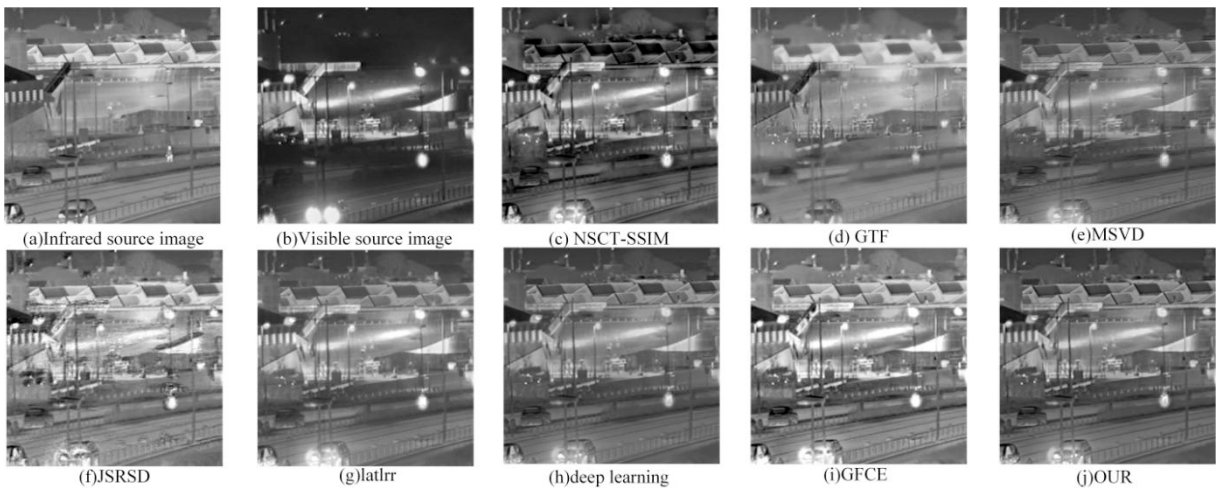


Fig. 6. Fused images of (d) test images.

(2) Objective evaluation

Three Q-series objective evaluation indicators ($Q^{AB/F}$ [14,17], Q^e [15,17], and Q^p [16,17]) are used to prove the superiority of the algorithm. Table.1-Table.4, respectively, give the objective evaluation results of the proposed algorithm and the remaining seven advanced fusion algorithms on four infrared visible-light images. It can be seen from the above table that the fusion evaluation index value of the proposed method is the highest. For the group (a) images, $Q^{AB/F}$, Q^e , and Q^p are 0.7113, 0.5627, and 0.7464, respectively. Compared with the deep learning fusion algorithm, the proposed method improves on the three evaluation indicators by 1.77%, 1.57%, and 2.41%, respectively. For the group (b) images, $Q^{AB/F}$, Q^e , and Q^p are 0.7276, 0.4843, and 0.7403, respectively. Compared with the deep learning fusion algorithm, the proposed method improves on the three evaluation indicators by 3.38%, 2.09%, and 3.26%, respectively. For the group (c) images, $Q^{AB/F}$, Q^e , and Q^p are 0.5999, 0.4775, and 0.5594, respectively. Compared with the deep learning fusion algorithm, the proposed method improves on the three evaluation indicators by 6.84%, 0.56%, and 6.63%, respectively. For the group (d) images, $Q^{AB/F}$, Q^e , and Q^p are 0.6138, 0.4891, and 0.5951, respectively. Compared with the deep learning fusion algorithm, the proposed method improves on the three evaluation indicators by 8.71%, 5.87%, and 6.76%, respectively. In summary, it can be seen from the objective evaluation indicators that our algorithm is superior to others.

Table 1. The objective evaluation of (a) group images.

	$Q^{AB/F}$	Q^e	Q^p
NSCT-SSIM ^[7]	0.6416	0.4973	0.4720
GTF ^[8]	0.5397	0.4158	0.3890
MSVD ^[9]	0.5792	0.5244	0.5977
JSRSD ^[10]	0.5227	0.4497	0.4953
LatIrr ^[11]	0.6084	0.4683	0.6176
Deep learning ^[12]	0.6989	0.5540	0.7288
GFCE ^[13]	0.5096	0.4352	0.4578
OUR	0.7113	0.5627	0.7464

Table 2. The objective evaluation of (b) group images.

	$Q^{AB/F}$	Q^e	Q^p
NSCT-SSIM ^[7]	0.6973	0.4499	0.6416
GTF ^[8]	0.4928	0.3588	0.2969
MSVD ^[9]	0.4648	0.4445	0.4133
JSRSD ^[10]	0.5031	0.3767	0.4034
LatIrr ^[11]	0.6063	0.4079	0.5920
Deep learning ^[12]	0.7038	0.4744	0.7169
GFCE ^[13]	0.5746	0.4591	0.5604
OUR	0.7276	0.4843	0.7403

Table 3. The objective evaluation of (c) group images.

	$Q^{AB/F}$	Q^e	Q^p
NSCT-SSIM ^[7]	0.5347	0.3826	0.3266
GTF ^[8]	0.2348	0.2104	0.1822
MSVD ^[9]	0.3659	0.4189	0.4573
JSRSD ^[10]	0.5439	0.3677	0.3739
LatIrr ^[11]	0.5814	0.3981	0.4655
Deep learning ^[12]	0.5615	0.4319	0.5246
GFCE ^[13]	0.4055	0.3992	0.3440
OUR	0.5999	0.4775	0.5594

Table 4. The objective evaluation of (d) group images.

	$Q^{AB/F}$	Q^e	Q^p
NSCT-SSIM ^[7]	0.6021	0.4441	0.4599
GTF ^[8]	0.3549	0.3038	0.2576
MSVD ^[9]	0.4120	0.4135	0.4012
JSRSD ^[10]	0.4485	0.3485	0.2817
LatIrr ^[11]	0.5359	0.3822	0.4763
Deep learning ^[12]	0.5646	0.4620	0.5574
GFCE ^[13]	0.5996	0.4578	0.5089
OUR	0.6138	0.4891	0.5951

5. Conclusion

We propose an infrared-visible image fusion algorithm via local convolution dictionary learning. Our method uses a standard block-based tool in the sparse domain to solve the global convolutional sparse coding problem. The experimental results show that learned convolution dictionary better expresses the global information of source images and retains local details.

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